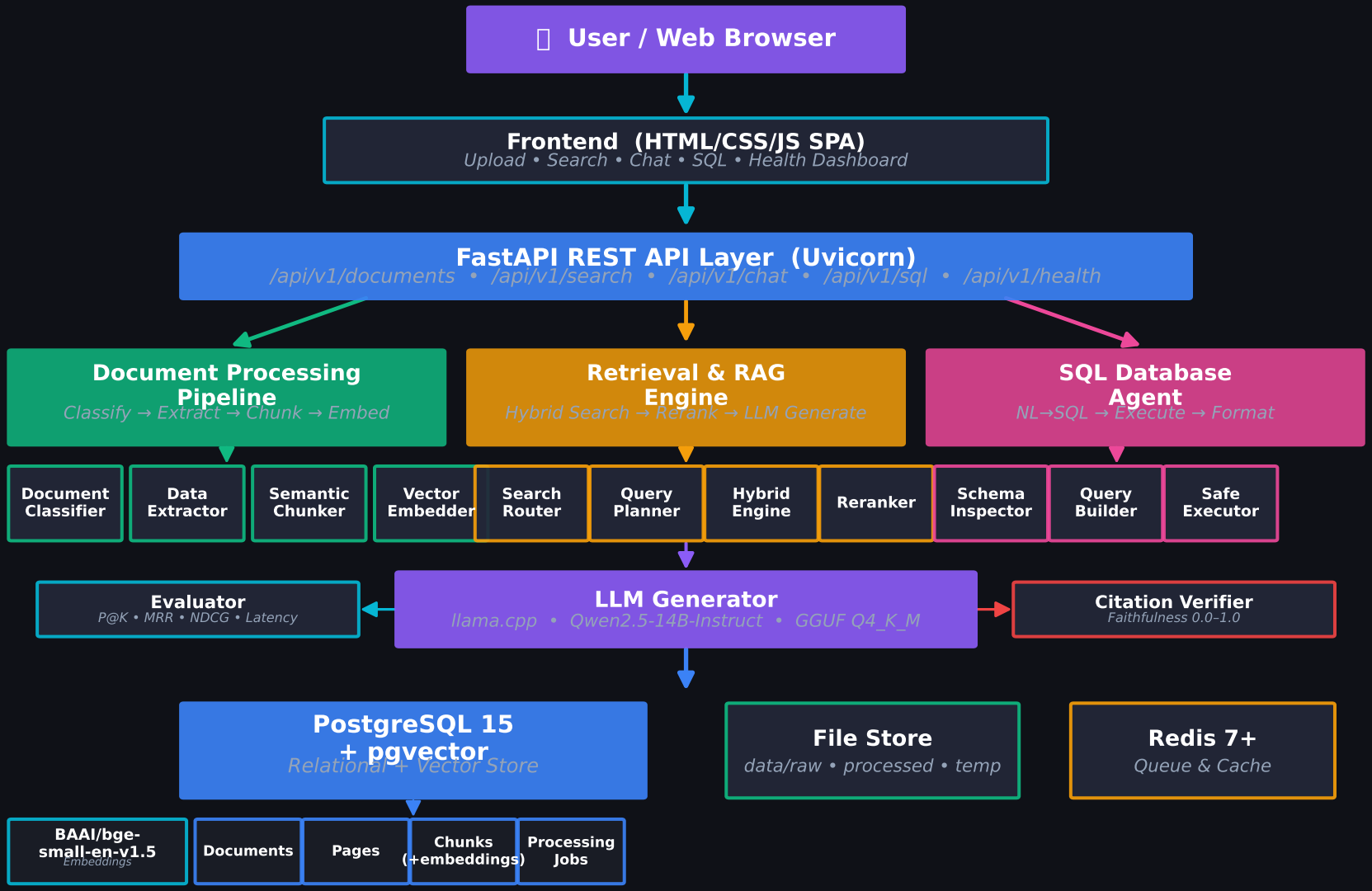
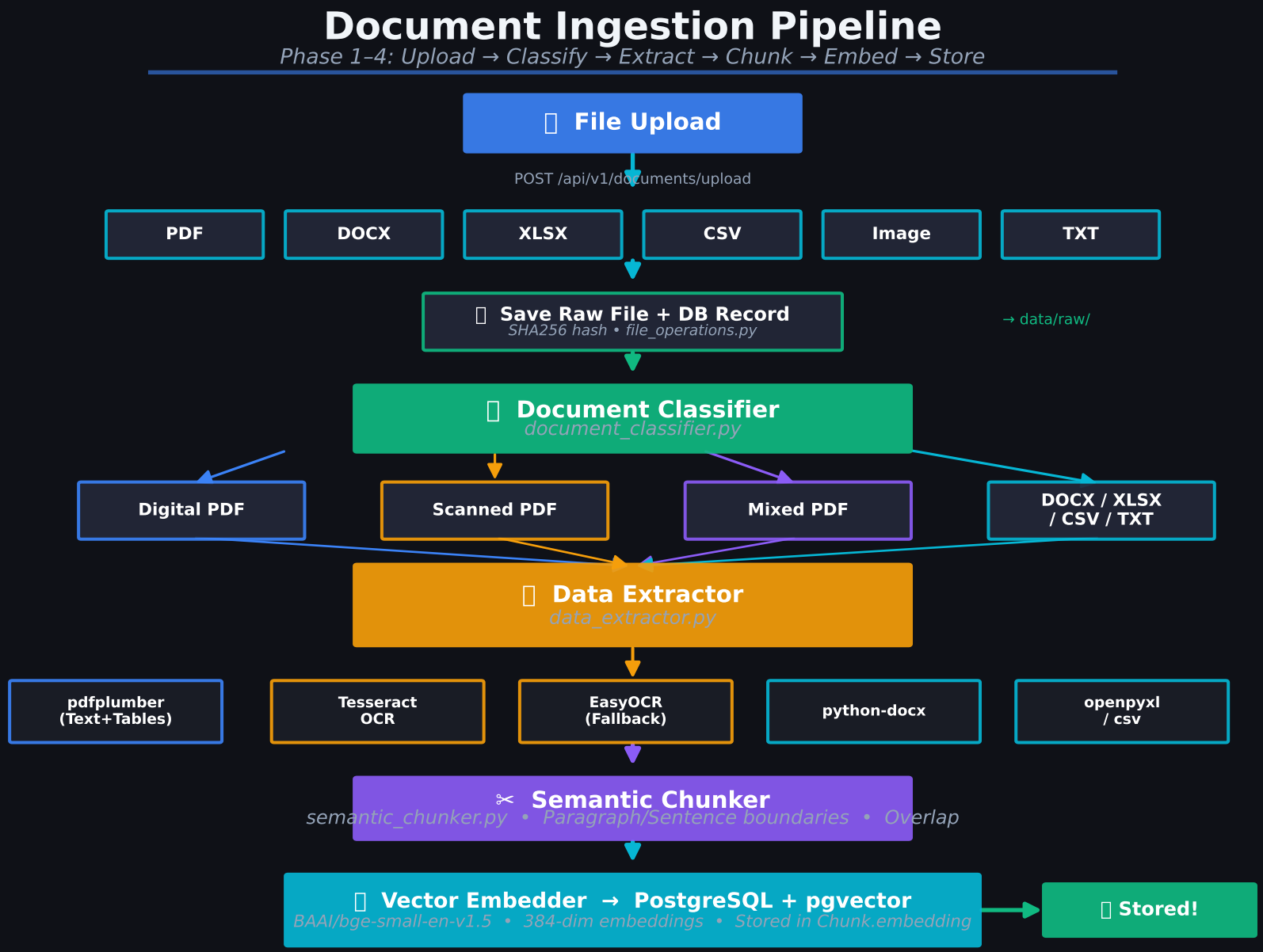
DocuRAG — High-Level System Architecture

Enterprise AI Document Intelligence & RAG System

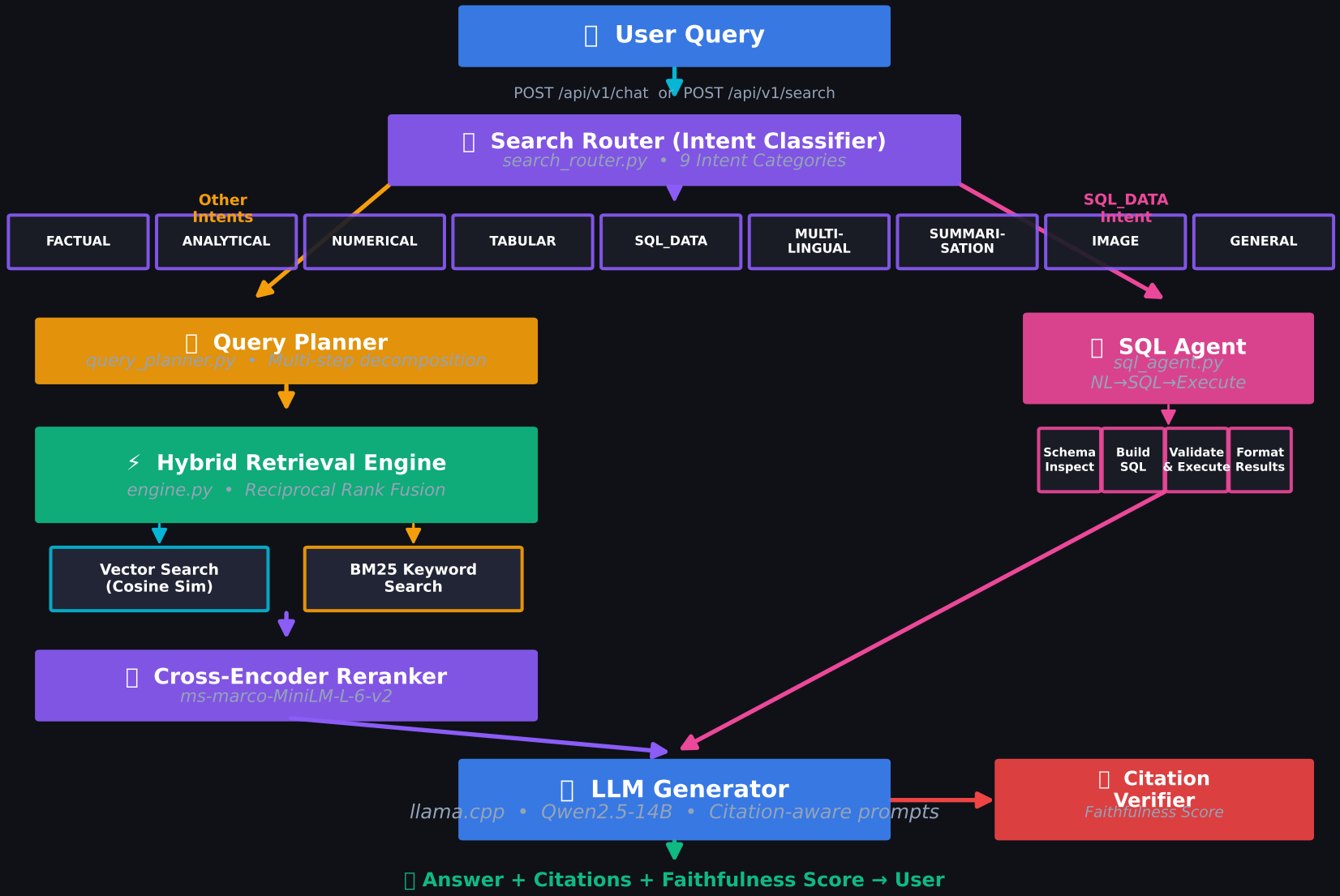


DocuRAG v1.0 — CPU-Optimised, Open-Source Enterprise Document Intelligence


Document Ingestion Pipeline
Phase 1–4: Upload → Classify → Extract → Chunk → Embed → Store
File Upload
POST /api/v1/documents/upload
PDF
DOCX
XLSX
CSV
Image
TXT
Save Raw File + DB Record
SHA256 hash • file_operations.py
→ data/raw/
Document Classifier
document_classifier.py
Digital PDF
Scanned PDF
Mixed PDF
DOCX / XLSX / CSV / TXT
Data Extractor
data_extractor.py
pdfplumber (Text+Tables)
Tesseract OCR
EasyOCR (Fallback)
python-docx
openpyxl / csv
Semantic Chunker
semantic_chunker.py • Paragraph/Sentence boundaries • Overlap
Vector Embedder → PostgreSQL + pgvector
BAAI/bge-small-en-v1.5 • 384-dim embeddings • Stored in Chunk.embedding
Stored!

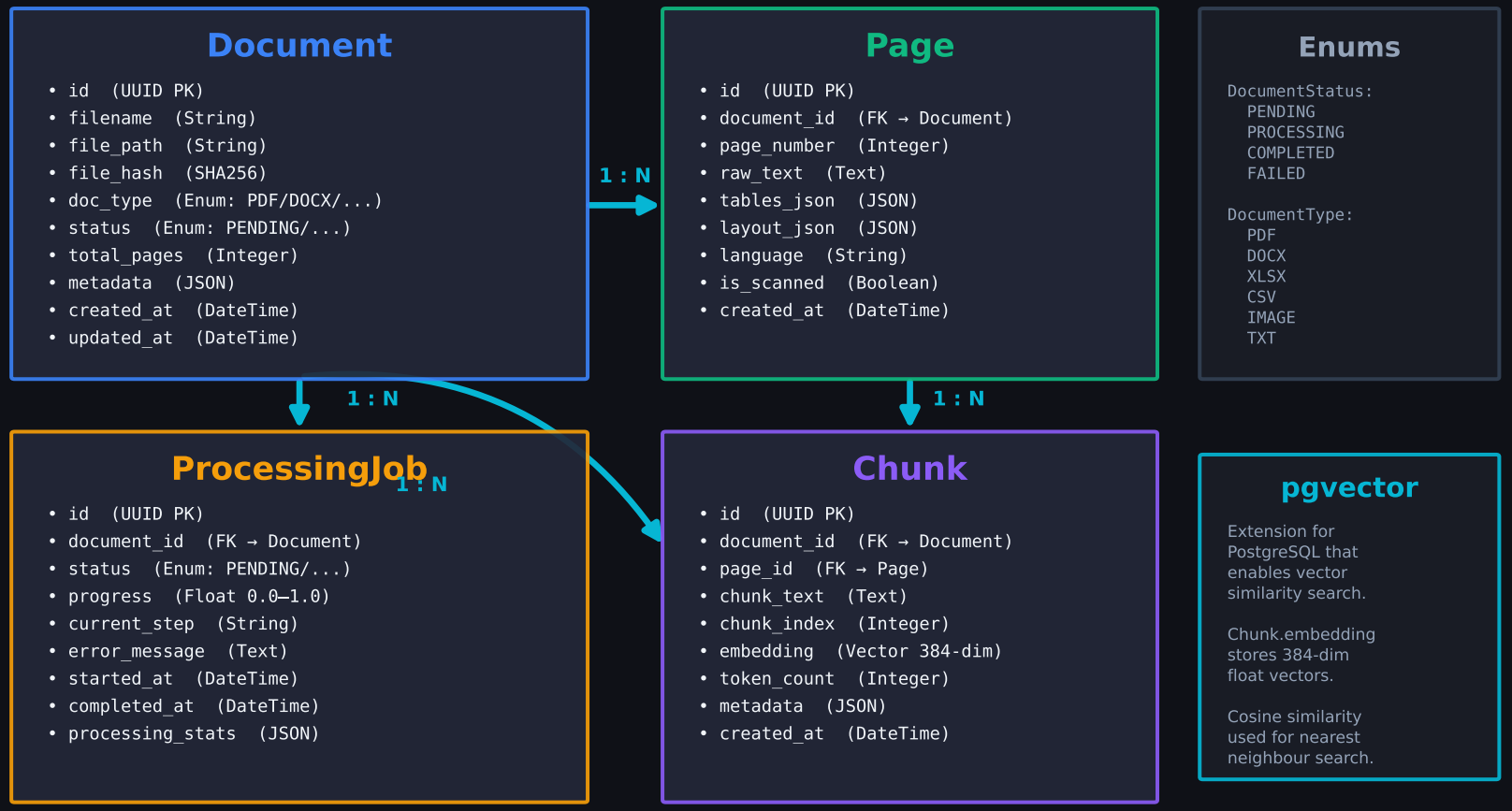
Retrieval & RAG Generation Pipeline

Phase 5–9: Query → Route → Retrieve → Rerank → Generate → Cite



Database Models & Data Relationships

SQLAlchemy 2.0 (Async) + PostgreSQL 15 + pgvector



All models defined in src/database_models/ • Migrations via Alembic • Async sessions via SQLAlchemy 2.0

API Layer & Frontend Architecture

FastAPI REST Endpoints + Single-Page Application

FastAPI REST API — /api/v1/

/health

```
GET /health  → System status
```

/documents

```
POST   /upload    → Ingest file
GET    /          → List all docs
GET    /{id}      → Doc details
DELETE /{id}      → Remove doc
```

/search

```
POST /          → Hybrid search
GET  /stats      → Index stats
```

/chat

```
POST /  → RAG-powered chat
```

/sql

```
POST   /connect         → Register DB
DELETE /connect/{label} → Deregister
GET    /connections     → List DBs
GET    /schema/{label}  → Inspect
POST   /query           → NL→SQL
```

/eval

```
POST /run  → Quality benchmarks
GET  /results  → Eval metrics
```

JSON Request/Response • CORS enabled • Swagger /docs • ReDoc /redoc

HTTP / JSON

Frontend — Single Page Application

Upload

Drag & drop
file upload
progress bar

Search

Hybrid search
results with
citations

Chat

Conversational
RAG with
faithfulness

SQL

Connect DBs
NL query
schema inspect

Health

System metrics
CPU/Memory
model status

`index.html + css/index.css + js/app.js`

Dark Theme • Glassmorphism • Responsive • Served as Static Files by FastAPI

Project File Tree & Module Map

Complete directory structure with module responsibilities

```
estate/  (DocuRAG Root)
├── application_configuration/
│   ├── environment_settings.py     Pydantic Settings (.env)
│   └── logger_setup.py             Structlog / logging config
├── src/
│   ├── main_application.py         FastAPI app entrypoint
│   ├── api/
│   │   ├── v1_router.py            API v1 router aggregator
│   │   └── routes/
│   │       ├── health_routes.py    GET /health
│   │       ├── document_routes.py  CRUD + upload endpoints
│   │       ├── search_routes.py    Hybrid search endpoints
│   │       ├── chat_routes.py      RAG chat endpoint
│   │       └── sql_routes.py       SQL agent endpoints
│   ├── database_models/
│   │   ├── database_connection.py  Async engine + sessions
│   │   ├── shared_enums.py         Status & Type enums
│   │   ├── document_model.py       Document ORM model
│   │   ├── page_model.py           Page ORM model
│   │   ├── chunk_model.py          Chunk + embedding ORM
│   │   └── processing_job_model.py Job tracking ORM
│   ├── document_processing/
│   │   ├── ingestion_pipeline.py   Orchestrator
│   │   ├── document_classifier.py  Digital/Scanned/Mixed
│   │   ├── data_extractor.py       Multi-format extraction
│   │   ├── semantic_chunker.py     Boundary-aware chunking
│   │   ├── vector_embedder.py      BGE embedding + store
│   │   └── processing_schemas.py   Pydantic DTOs
│   ├── retrieval/
│   │   ├── engine.py               Hybrid retrieval + RRF
│   │   ├── retriever.py            Vector + BM25 retrievers
│   │   ├── reranker.py             Cross-encoder reranker
│   │   ├── search_router.py        9-intent classifier
│   │   ├── query_planner.py        Multi-step decomposition
│   │   ├── search_service.py       High-level orchestrator
│   │   ├── sql_agent.py            NL→SQL agent
│   │   ├── citation_verifier.py    Faithfulness scoring
│   │   └── evaluator.py            P@K, MRR, NDCG metrics
│   ├── llm/
│   │   └── generator.py            llama.cpp + extractive
│   └── shared_utilities/
│       ├── file_operations.py      File I/O + hashing
│       └── text_cleaning.py        Text normalization
├── frontend/
│   ├── index.html                  SPA shell
│   ├── css/index.css               Dark glassmorphism UI
│   └── js/app.js                   Frontend logic
├── alembic/                        DB migrations
├── tests/                          Pytest suite
├── scripts/                        Setup scripts
├── data/  (raw/ processed/ temp/)  Runtime data
├── models/                         GGUF LLM weights
└── logs/                           Runtime logs
```